

Gate Stack Engineering of 3D Oxide Channel FeNOR Memory with High-Speed and Reliability

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Abstract—In this work, we systematically engineer the gate stack (MFS, MFIS, MIFS, and MIFIS) in 3D vertical oxide-channel FeNORs to optimize memory characteristics. The interlayer-free MFS device is demonstrated to provide the optimal overall performance, achieving the fastest operation speed of 0.25 μ s and highest endurance of $>10^{11}$, while maintaining a practical 1V memory window and robust retention. Characterizations of interface and oxide traps reveal that suppressed charge trapping in MFS is the critical mechanism for enhanced reliability. Our work presents a viable strategy for realizing fast and reliable 3D FE memory.

Keywords—3D FeNOR, oxide channel, interlayer-free, charge trapping, high endurance

I. INTRODUCTION

The deployment of large language models (LLMs) at the edge has attracted significant interest due to its low latency, reduced cost, and enhanced privacy [1]. However, this trend faces a severe memory bound, imposing stringent demands on storage solutions for higher bandwidth, density, and power efficiency [2]. Recently, 3D NOR-type ferroelectric field-effect transistor (FeFET), or 3D FeNOR, has emerged as one of the most promising candidates to address these challenges [3]. Its 3D NOR vertical stacked architecture provides high density and random access capability, meanwhile the high speed, low power consumption, and scalability of FeFET guarantees balanced performance. However, the poor reliability, especially endurance, remains a key concern for 3D FeNOR. Introducing oxide channels in 3D FeNOR is expected to address the critical endurance challenge [4], yet typically result in compromises in speed and MW [5]. Gate stack engineering emerges as a key strategy to address this trade off. Unfortunately, the impact of different gate stacks, particularly the interlayers (ILs), remains controversial. While some reports suggest a channel IL is indispensable for reliability [6], many high-performance oxide channel FeFETs feature an IL-free structure [7, 8]. Additionally, a gate IL can effectively enhance the memory window (MW), yet the impact on speed and reliability is so far unclear. Therefore, in-depth research into these gate stack designs is required to achieve comprehensive optimization in MW, speed, and endurance.

In this work, we experimentally fabricate and compare 3D FeNORs with distinct gate stacks (MFS, MFIS, MIFS, MIFIS). We demonstrate that the IL-free MFS design delivers superior overall performance. Through this gate stack engineering, the 3D FeNOR achieves a fast operation speed of 250 ns, excellent endurance of $>10^{11}$ cycles, reasonable MW of 1 V, and robust retention. Furthermore, comprehensive characterization of charge trapping proves that the reduced trapped charges in MFS device is the key to speed and reliability improvement.

II. RESULTS AND DISCUSSION

A. Fabrication and Memory Window of 3D FeNOR

Fig. 1 (a) illustrates the structure of 3D oxide channel FeNOR, where the drain current flows horizontally around the cylinder. 3-tier-stacked FeNORs are fabricated followed the process flow in Fig. 1 (d). After depositing the gate and spacer layers, vertical trenches are etched, followed by the growth of various gate stacks via atomic layer deposition (ALD). A 2 nm Al_2O_3 is inserted between FE layer and oxide channel or between FE layer and gate as the channel (Ch.IL) or gate interlayer (G.IL), respectively. An additional TiN layer is applied to induce ferroelectricity in rapid thermal annealing (RTA), then removed. 8 nm InGaO (IGO) is grown by ALD as oxide channel, then the source and drain are formed by lift-off techniques. The effective gate length (L_g) is scaled down to 100 nm, enabled by e-beam lithography. The cross-sectional HRTEM and the corresponding EDS images in Fig. 1 (b) show the 3-tier stacked structure, with smallest gate width of 10 nm. Clear interface between each layer is also demonstrated.

The I_d - V_g curves for 3D FeNORs with various gate stacks are shown in Fig. 2. All devices show typical counterclockwise hysteresis with $I_{on}/I_{off} > 10^5$. MFS and MFIS exhibit MW of 1 V and 0.8 V, respectively, while the MW increases to ~ 2 V for devices with G.IL. This MW enhancement is primarily caused by gate-side charge injection. The dependence of MW on L_g is investigated in Fig. 3. For all gate stacks, the MW increase as L_g shrinks from 1 μ m to 100 nm. This result is consistent with our previous work and is well explained in [9]. It is noteworthy that as L_g increases beyond 300 nm, G.IL has no significant effect on MW, which is then dominated by size effect.

B. Characterization of Charge Trapping in 3D FeNORs

To investigate the information of interface traps in various gate stack saps, we conducted multi-frequency C-V measurements on all the devices, and the results are presented in Fig. 4. AC signals from 1 kHz to 100 kHz are employed to test the gate capacitance C_{gs} . Since the response of interface traps is non-transient, the frequency dispersion of the C-V curves between high and low frequencies can qualitatively reflect the amount of interface traps between FE layer and oxide channel. Compared to MFIS, MFS exhibits smaller frequency dispersion in the depletion region (Fig. 4 (a)), indicating that the additional Ch.IL rather degrades the interface quality between the FE and oxide channel. In addition, introducing G.IL also increases the number of interface traps.

To provide more insights, we further characterize the devices with the double quasi-static CV (DQSCV) method, as depicted in Fig. 5. The source and drain are connected to monitor the trapping current, and the positive-up-negative-down (PUND) signal is applied to eliminate the impact of non-ideal leakage current. Fig. 6. displays the measured current

response (I_{SD}) and the corresponding trapped charges (Q_{trap}) obtained by integrating I_{SD} over time. Compared to MFIS, MFS exhibits a smaller Q_{trap} of $10.77 \mu C/cm^2$, suggesting that the IL-free structure effectively suppresses charge trapping from channel to gate stack. Both MIFS and MIFIS exhibit a significantly increased Q_{trap} of $\sim 120 \mu C/cm^2$, which is induced by the massive trapped charges at G.IL. Moreover, unlike MIFS and MIFIS, in devices without G.IL, current response occurs only at negative V_g , while I_{SD} is zero at positive V_g . This is due to the inherent n-type nature and the difference in the amount of trapped charge. In MFS and MFIS, the trapped charges at positive voltages are all from the channel. Conversely, in MIFS and MIFIS, the electrons in channel cannot fully compensate the polarization and gate-side trapped charges, thus resulting in current response at source and drain.

Fig. 7 details the charge trapping behavior in various gate stacks. Compared to MFIS, MFS has fewer interface and oxide traps, hence charge trapping is suppressed during program (PGM), resulting in less de-trapped charges in erase (ERS). Whereas in MIFS, the extra gate-side charge injection induces much more trapped charges at channel side, thereby triggering the I_{SD} response. Compared to other gate stacks, the reduced trapped charge in the IL-free structure may enhance the reliability of 3D FeNORs.

C. Speed and Reliability of 3D FeNORs

The program and erase speeds for various gate stacks are measured in Fig. 8. The operation voltages are set to 3.75V for MFS and MFIS, and to 5V for MIFS and MIFIS for fair comparison. The V_{th} shift (ΔV_{th}) is measured after pulses with increasing width, and the switch time is determined as the ΔV_{th} reaches 90% of the maximum MW. Among all gate stacks, MFS demonstrates the fastest operation speed of 0.25 μs . In MFIS, the operation speed is slowed down to 1.5 μs . This finding agrees with other work [6] and may be attributed to the increased charge trapping from channel. The operation speed is further slowed to 2 μs and 5 μs in MIFS and MIFIS, respectively. This is because in devices with G.IL, programming and erasing are accompanied by gate charge injection, which requires a longer time compared to polarization switching.

We then conduct endurance measurements on the 3D FeNORs, with results shown in Fig. 9. As expected, MFS demonstrated the highest endurance of $>10^{11}$ cycles, with MW showing little degradation during cycling. On the contrary, MFIS suffers early breakdown after 10^7 cycles, while the MW rapidly degrades after 10^6 cycles in both MIFS and MIFIS. This endurance comparison perfectly aligns with the characterization of trapped charges, where the suppressed charge trapping behavior in MFS gate stack effectively optimizes the endurance. Retention measurement is conducted for 3D FeNORs at room temperature, as displayed in Fig. 10. All gate stacks exhibit stable V_{th} for up to 4000 s, which can be linearly extrapolated to 10 years. Nevertheless, the V_{th} shift is still larger for devices with G.IL than those without G.IL, which requires further optimization.

The key memory performance of the 3D FeNORs is benchmarked with other 3D oxide channel FeFETs in Table. 1 [10]. The IL-free MFS design in this work demonstrates superior overall performance, including the highest endurance, fastest operation speed, lowest voltage with reasonable MW and robust retention.

III. CONCLUSION

This study presents a comparative investigation of various gate stacks in 3D oxide channel FeNORs. Our results reveal that an IL-free MFS structure provides a superior balance of memory characteristics. This optimal design yields a switch time of 0.25 μs , excellent endurance of $>10^{11}$ cycles, and robust retention. Through comprehensive characterization, these enhancements are attributed to the mitigated charge trapping at FE/oxide channel interface. These findings establish the IL-free design as a promising pathway toward next-generation high-performance 3D FE memories.

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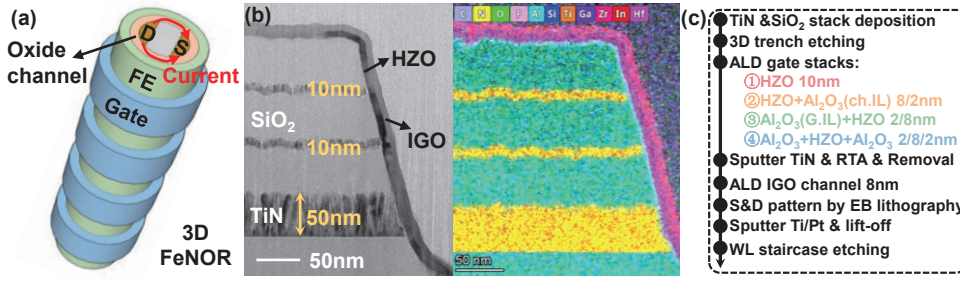


Fig. 1 (a) Structure schematic of 3D FeNORs with oxide channel. (b) Cross-sectional HRTEM image and the corresponding EDS images of the fabricated 3-tiers stacked 3D FeNORs. Clear interfaces and a minimum gate width of 10nm are demonstrated. (c) Process flow of 3D FeNORs. Four gate stacks are included for comparison.

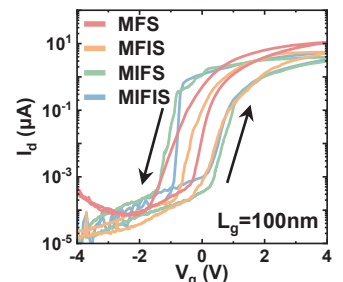


Fig. 2 I_d - V_g curves for the 3D FeNORs with various gate stacks, with MW ranging from 1V to 2V.

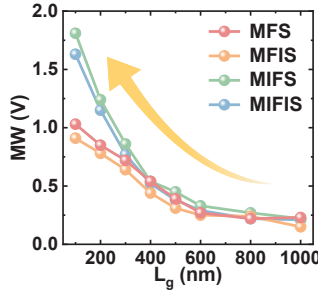


Fig. 3 Dependence of MW on L_g . For all the four gate stacks, the MW increases as L_g shrinks.

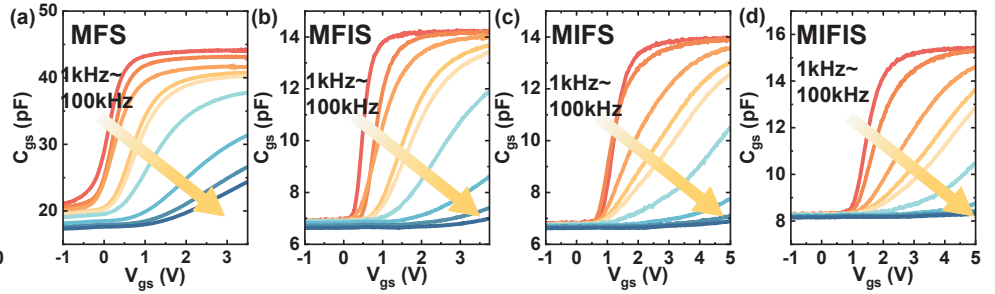


Fig. 4 Multi-frequency C-V measurements for 3D FeNORs with (a) MFS, (b) MFIS, (c) MIFS, and (d) MIFIS gate stack, with ac frequency from 1kHz to 100kHz. The weaker frequency dispersion in MFS stack indicates the fewer interface traps between HZO and oxide channel compared to other gate stacks.

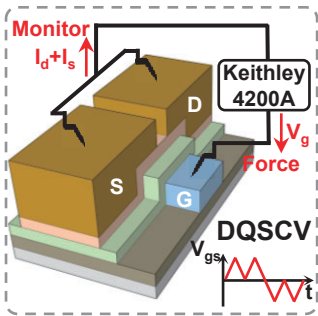


Fig. 5 Schematic of the DQSCV test. Drain and source are connected to measure the trapping current.

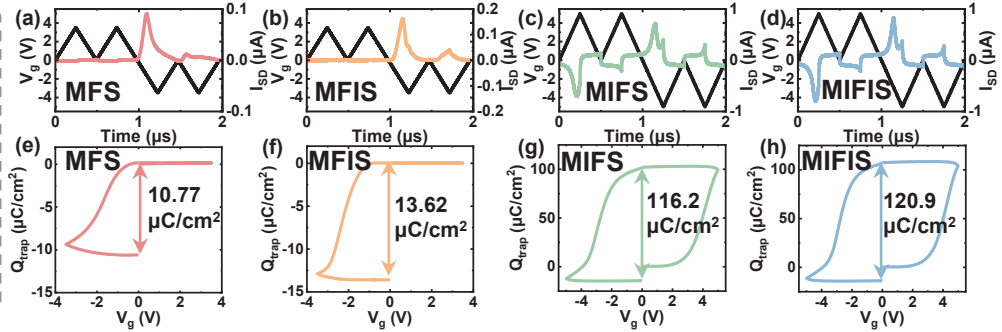


Fig. 6 (a-d) Measured I_{SD} and (e-h) corresponding Q_{trap} for the four gate stacks. Q_{trap} is obtained by integrating the effective I_{SD} over time. MIFS shows a larger Q_{trap} compared to MFS, while MIFS and MIFIS demonstrates significantly higher Q_{trap} of 116.2 and 120.9 $\mu C/cm^2$, which is induced by gate-side charge injection.

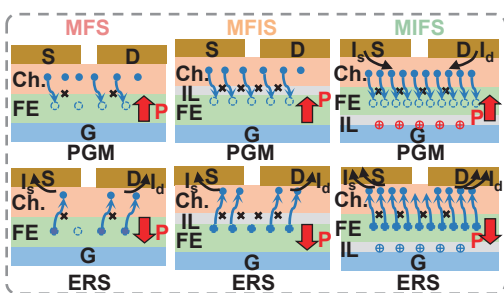


Fig. 7 Charge trapping mechanisms in different gate stacks. Channel IL leads to more interface and oxide traps, while gate IL further significantly increases trapped charges

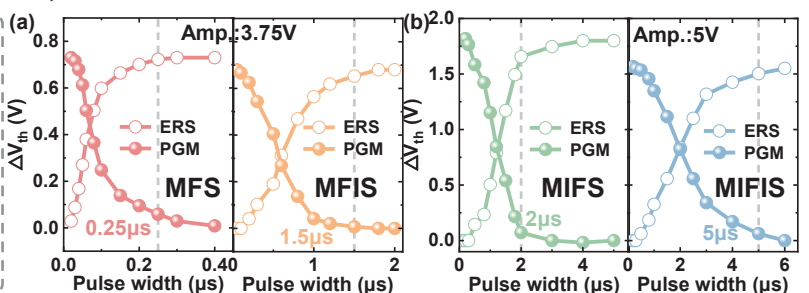


Fig. 8 Measured program and erase speed for (a) MFS/MFIS and (b) MIFS/MIFIS gate stack. MFS demonstrates fastest operation of 0.25 μs while all gate stacks with IL show much slower speed of $>1 \mu s$.

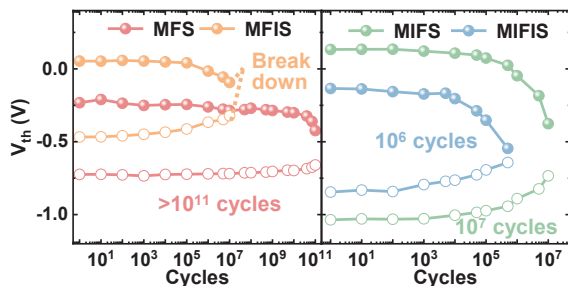


Fig. 9 Endurance measurements for 3D FeNORs. MFS gate stack exhibits highest endurance of $>10^{11}$ cycles, while other gate stacks show endurance less than 10^7 cycles.

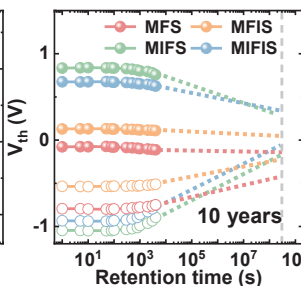


Fig. 10 Retention for 3D FeNORs. All gate stacks show stable V_{th} for 4000 s, extrapolated to 10 years.

Table. 1 Memory performance comparison for 3D oxide channel FeFETs

	[5]	[10]	In this work			
Struct.	MFS	MFS	MFS	MFIS	MIFS	MIFIS
MW	0.7V	2.0V	1.0V	0.8V	2.0V	1.6V
Speed	-	1 μs	0.25 μs	1.5 μs	2 μs	5 μs
Volt.	7V	7V	3.75V	3.75V	5V	5V
Endurance	10 ⁴	10 ⁸	$>10^{11}$	10 ⁷	10 ⁷	10 ⁶
Retention	10 ³ s	-	10yrs	10yrs	10yrs	10yrs